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```
#exercise 8-----------------------------------------------

#q1
student_data %>% filter(final_exam_mark > 80)
```

```
> #q1
> student_data %>% filter(final_exam_mark > 80)
  student_id  student_name gender coursework_mark final_exam_mark
1       S007   David Jones   Male              64              84
2       S032  Joseph Smith   Male              94              82
3       S044 Jessica Brown Female              77              84
```

```
#q2
student_data %>% arrange(coursework_mark)
```

```
> #q2
> student_data %>% arrange(coursework_mark)
   student_id      student_name gender coursework_mark final_exam_mark
1        S028    Charles Garcia   Male              34              39
2        S001  Elizabeth Miller Female              40              41
3        S006   Sarah Rodriguez Female              41              35
4        S036       Karen Jones Female              41              45
5        S038  Elizabeth Miller Female              41              34
6        S037       David Davis   Male              42              38
7        S030  Barbara Williams Female              43              55
8        S033    Richard Miller   Male              43              42
9        S040      Sarah Miller Female              43              51
10       S043 Patricia Williams Female              46              38
11       S013        John Smith   Male              50              53
12       S047   Linda Rodriguez Female              50              54
13       S015  Elizabeth Garcia Female              51              51
14       S022     Robert Miller   Male              51              54
15       S012       Mary Garcia Female              52              55
16       S011      David Miller   Male              55              49
17       S019    Susan Williams Female              55              44
18       S004     William Davis   Male              56              62
19       S039    Linda Williams Female              57              70
20       S016  Robert Rodriguez   Male              58              55
21       S002    William Garcia   Male              59              63
22       S027   Charles Johnson   Male              59              48
23       S017      Linda Garcia Female              61              74
24       S035    Barbara Miller Female              61              54
25       S049       Linda Davis Female              61              53
26       S048    Jessica Garcia Female              63              54
27       S007       David Jones   Male              64              84
28       S023    David Martinez   Male              64              55
29       S021     Robert Miller   Male              66              62
30       S026   Charles Johnson   Male              66              57
31       S010    Jessica Garcia Female              67              65
32       S014        John Brown   Male              67              58
33       S020  Joseph Rodriguez   Male              69              62
34       S009    Michael Garcia   Male              70              64
35       S046        Mary Davis Female              70              60
36       S025   Joseph Williams   Male              72              73
37       S008   Thomas Martinez   Male              73              60
38       S050      Robert Brown   Male              73              57
39       S005        John Smith   Male              74              56
40       S041  Barbara Martinez Female              75              55
41       S031   Joseph Williams   Male              76              66
42       S034  Richard Martinez   Male              76              67
43       S044     Jessica Brown Female              77              84
44       S045       Susan Brown Female              77              77
45       S003     Sarah Johnson Female              79              64
46       S029       Linda Smith Female              82              68
47       S024     William Jones   Male              87              64
48       S018   Sarah Rodriguez Female              93              70
49       S032      Joseph Smith   Male              94              82
50       S042      Sarah Miller Female              97              73
>
```

```r
#q3
student_data %>% select(student_name, final_exam_mark)
```

```
> #q3
> student_data %>% select(student_name, final_exam_mark)
        student_name final_exam_mark
1   Elizabeth Miller              41
2     William Garcia              63
3      Sarah Johnson              64
4      William Davis              62
5         John Smith              56
6    Sarah Rodriguez              35
7        David Jones              84
8    Thomas Martinez              60
9     Michael Garcia              64
10    Jessica Garcia              65
11      David Miller              49
12       Mary Garcia              55
13        John Smith              53
14        John Brown              58
15  Elizabeth Garcia              51
16  Robert Rodriguez              55
17      Linda Garcia              74
18   Sarah Rodriguez              70
19    Susan Williams              44
20  Joseph Rodriguez              62
21     Robert Miller              62
22     Robert Miller              54
23    David Martinez              55
24     William Jones              64
25   Joseph Williams              73
26   Charles Johnson              57
27   Charles Johnson              48
28    Charles Garcia              39
29       Linda Smith              68
30  Barbara Williams              55
31   Joseph Williams              66
32      Joseph Smith              82
33    Richard Miller              42
34  Richard Martinez              67
35    Barbara Miller              54
36       Karen Jones              45
37       David Davis              38
38  Elizabeth Miller              34
```

```
34  Richard Martinez                 67
35    Barbara Miller                 54
36       Karen Jones                 45
37       David Davis                 38
38  Elizabeth Miller                 34
39    Linda Williams                 70
40      Sarah Miller                 51
41  Barbara Martinez                 55
42      Sarah Miller                 73
43 Patricia Williams                 38
44     Jessica Brown                 84
45       Susan Brown                 77
46        Mary Davis                 60
47   Linda Rodriguez                 54
48    Jessica Garcia                 54
49       Linda Davis                 53
50      Robert Brown                 57
>
```

```r
#q4
nrow(student_data)
ncol(student_data)
```

```
> #q4
> nrow(student_data)
[1] 50
>
>
>
> ncol(student_data)
[1] 5
```

```
#q5
str(student_data)
summary(student_data)
head(student_data)
View(student_data)
```

```
> #q5
> str(student_data)
'data.frame':   50 obs. of  5 variables:
 $ student_id     : chr  "S001" "S002" "S003" "S004" ...
 $ student_name   : chr  "Elizabeth Miller" "William Garcia" "Sarah Johnson" "William Davis" ...
 $ gender         : chr  "Female" "Male" "Female" "Male" ...
 $ coursework_mark: int  40 59 79 56 74 41 64 73 70 67 ...
 $ final_exam_mark: int  41 63 64 62 56 35 84 60 64 65 ...
> summary(student_data)
     student_id    student_name       gender   coursework_mark final_exam_mark
 Length   :50   Length   :50    Length   :50   Min.   :34.00   Min.   :34.00
 N.unique :50   N.unique :42    N.unique : 2   1st Qu.:51.00   1st Qu.:51.50
 N.blank  : 0   N.blank  : 0    N.blank  : 0   Median :62.00   Median :56.50
 Min.nchar: 4   Min.nchar:10    Min.nchar: 4   Mean   :62.42   Mean   :57.88
 Max.nchar: 4   Max.nchar:17    Max.nchar: 6   3rd Qu.:73.00   3rd Qu.:64.75
                                               Max.   :97.00   Max.   :84.00
> head(student_data)
  student_id     student_name gender coursework_mark final_exam_mark
1       S001 Elizabeth Miller Female             40              41
2       S002   William Garcia   Male             59              63
3       S003    Sarah Johnson Female             79              64
4       S004    William Davis   Male             56              62
5       S005       John Smith   Male             74              56
6       S006  Sarah Rodriguez Female             41              35
> View(student_data)
>
```

Filter

	student_id	student_name	gender	coursework_mark	final_exam_mark
1	S001	Elizabeth Miller	Female	40	41
2	S002	William Garcia	Male	59	63
3	S003	Sarah Johnson	Female	79	64
4	S004	William Davis	Male	56	62
5	S005	John Smith	Male	74	56
6	S006	Sarah Rodriguez	Female	41	35
7	S007	David Jones	Male	64	84
8	S008	Thomas Martinez	Male	73	60
9	S009	Michael Garcia	Male	70	64
10	S010	Jessica Garcia	Female	67	65
11	S011	David Miller	Male	55	49
12	S012	Mary Garcia	Female	52	55
13	S013	John Smith	Male	50	53
14	S014	John Brown	Male	67	58
15	S015	Elizabeth Garcia	Female	51	51
16	S016	Robert Rodriguez	Male	58	55
17	S017	Linda Garcia	Female	61	74
18	S018	Sarah Rodriguez	Female	93	70
19	S019	Susan Williams	Female	55	44
20	S020	Joseph Rodriguez	Male	69	62

Showing 1 to 20 of 50 entries, 5 total columns

Explanation Q5)

1. Str(student_data): Shows the structure of the dataset, such as number of rows or columns, each column's name, and its data type. Quickly checks if columns have the correct data type.

2. Summary(student_data): Shows summary statistics for each column, such as min, max, mean, median, and quartiles for numeric columns. Quickly spots outliers, spread, and missing values.

3. Head(student_data): Shows the first 6 rows of the dataset such as small preview of the actual data. Confirms the data loaded correctly without printing everything.

4. View(student_data): Opens the dataset in a spreadsheet style viewer such as full dataset in a scrollable table. Easy to visually browse and check specific values.

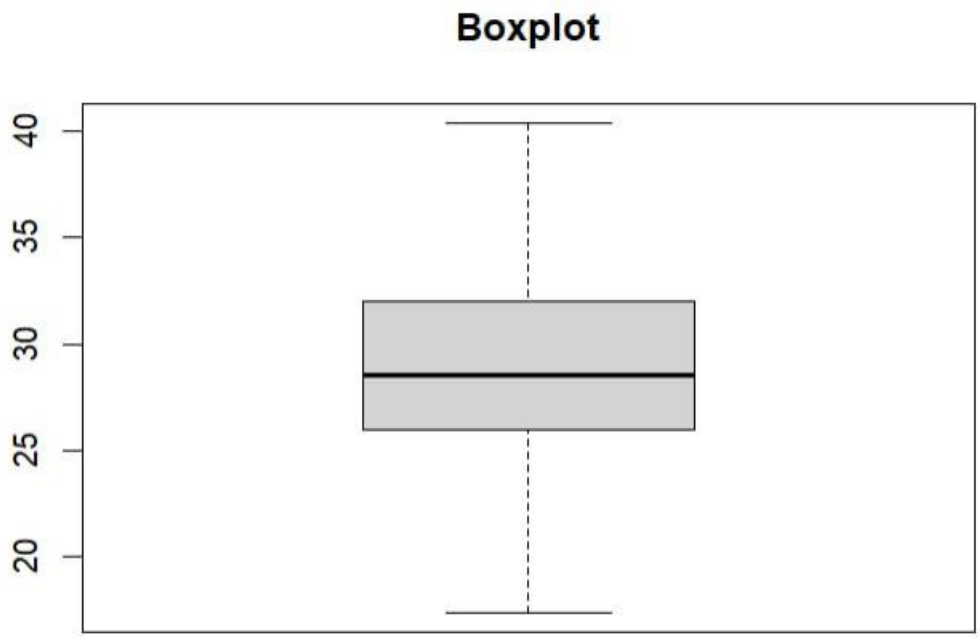
```r
#q6
# Step 1: Calculate quartiles and IQR
Q1 <- quantile(student_data$final_exam_mark, 0.25)
Q3 <- quantile(student_data$final_exam_mark, 0.75)
IQR_value <- IQR(student_data$final_exam_mark)

# Step 2: Calculate lower and upper bounds
lower_bound <- Q1 - 1.5 * IQR_value
upper_bound <- Q3 + 1.5 * IQR_value

# Step 3: Identify the outliers
outliers <- student_data %>% filter(final_exam_mark < lower_bound | final_exam_mark > upper_bound)

outliers
nrow(outliers)
```

```
> outliers <- student_data %>% filter(final_exam_mark < lower_bound | final_exam_mark >
> outliers
[1] student_id      student_name    gender          coursework_mark final_exam_mark
<0 rows> (or 0-length row.names)
> nrow(outliers)
[1] 0
>
```



Explanation Q6)

Outliers are values that are unusually high or low compared to the rest of the data. A common way to detect them is the IQR method any value below (Q1 - 1.5×IQR) or above (Q3 + 1.5×IQR) is considered an outlier.

Result from the boxplot : No points appeared above or below the whiskers of the boxplot, and the outliers table returned empty. This means no outliers were detected